

WBJEE -2026

MOCK TEST: MATHEMATICS

Total Marks: 100 | Time Allowed: 2 Hours

General Instructions

1. This paper contains 75 questions divided into three categories
2. **Category-I (Q1 - Q50):** 1 Mark each. Only one option is correct. Negative marking is -0.25 marks.
3. **Category-II (Q51 - Q65):** 2 Marks each. Only one option is correct. Negative marking is -0.5 marks.
4. **Category-III (Q66 - Q75):** 2 Marks each. One or more options may be correct. No negative marking. Partial marks are awarded for partially correct answers.

Category: I (Q: 1 to 50)

Category -1: carry 1 mark each and only one option is correct.

1. Let $f(x) = \sqrt{\log_{10} x^2}$. The set of all values of x for which $f(x)$ is real is

- a) $[1, \infty]$
- b) $[-1, 1]$
- c) $[-\infty, -1]$
- d) $[-\infty, -1] \cup [1, \infty]$

Ans: d)

2. If $\log_{\cos x} \tan x + \log_{\sin x} \cot x = 0$, then the most general solutions of x are

- a) $n\pi + \frac{\pi}{4}$; $n \in \mathbb{Z}$
- b) $2n\pi + \frac{\pi}{4}$; $n \in \mathbb{Z}$
- c) $2n\pi - \frac{3\pi}{4}$; $n \in \mathbb{Z}$
- d) $n\pi - \frac{\pi}{4}$; $n \in \mathbb{Z}$

Ans: b)

3. If x is a real number such that $x(x^2 + 1), -\frac{x^2}{2}, 6$ are three consecutive terms of an A.P. Then the next two consecutive terms of the A.P. are

- a) 14 , 6
- b) -2 , -10
- c) 14 , 22

d) -2 , 6

Ans: c)

4. Let a, b, c be real numbers such that $a + b + c < 0$ and the quadratic equation $ax^2 + bx + c = 0$ has imaginary roots, then

a) $a > 0, c > 0$

b) $a < 0, c > 0$

c) $a > 0, c < 0$

d) $a < 0, c < 0$

Ans: d)

5. If $x^2 + 9y^2 + 25z^2 = xyz \left(\frac{15}{x} + \frac{5}{y} + \frac{3}{z} \right)$ then x, y, z are in

a) A.P.

b) G.P.

c) H.P.

d) none

Ans: c)

6. Solution of the inequation $|z - 4| < |z - 2|$ is (where z is a Complex number)

a) $\operatorname{Re} z > 0$

b) $\operatorname{Re} z < 0$

c) $\operatorname{Re} z > 3$

d) $\operatorname{Re} z < 3$

Ans: c)

7. The argument of the complex number $z = 1 + i \tan \frac{2\pi}{5}$ is

a) $-\frac{2\pi}{5}$

b) $\frac{2\pi}{5}$

c) $-\frac{3\pi}{5}$

d) $\frac{3\pi}{5}$

Ans: a)

8. Out of 10 consonants and 4 vowels, how many words can be formed each containing 3 consonants and 2 vowels?

a) $\binom{10}{3} \cdot \binom{4}{3} \times 5!$

b) $\binom{10}{3} \times 5!$

c) $\binom{10}{3} \cdot \binom{4}{3}$

d) $\binom{10}{3}$

Ans: a)

9. If $f(x) = \sin x + \cos x$ and $g(x) = x^2 - 1$, then $g\{f(x)\}$ is invertible if

a) $-\frac{\pi}{2} \leq x \leq 0$

b) $-\frac{\pi}{2} \leq x \leq \pi$

c) $0 \leq x \leq \frac{\pi}{2}$

d) $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$

Ans: d)

10. The value of $\lim_{x \rightarrow 0} (1 + 3x)^{\frac{x+2}{x}}$ equal to

a) e^3

b) e^6

c) e^5

d) e

Ans: b)

11. If $2^x + 2^y = 2^{x+y}$ then the value of $\frac{dy}{dx}$ at (1,1) is

a) 0

b) -1

c) 1

d) 2

Ans: b)

12. Let $f(x) = x(x-1)(x-2)$, $x \in [0,4]$; then the point $x = c$ which satisfies Mean Value Theorem satisfies

a) $0 < c < 1$

- b) $c > 3$
 c) $0 < c < 1/2$
 d) $1 < c < 3$

Ans: d)

13. $\int \frac{x^5}{\sqrt{1+x^3}} dx =$

- a) $\frac{2}{3} \left[\frac{(1+x^3)^{\frac{3}{2}}}{3} - \sqrt{1+x^3} \right] + c$
 b) $\frac{1}{3} \left[\frac{(1+x^3)^{\frac{3}{2}}}{3} - \sqrt{1+x^3} \right] + c$
 c) $\frac{2}{3} \left[\frac{(1+x^3)^{\frac{3}{2}}}{3} + \sqrt{1+x^3} \right] + c$
 d) $\frac{1}{3} \left[\frac{(1+x^3)^{\frac{3}{2}}}{3} + \sqrt{1+x^3} \right] + c$

Ans: a)

14. If $\int \frac{4e^x + 6e^{-x}}{9e^x - 4e^{-x}} dx = Ax + B \log(9e^{2x} - 4) + C$, then find A, B & C.

- a) $A = \frac{5}{2}, B = \frac{31}{36}, C = -\frac{3}{2} \log 3 + c$
 b) $A = \frac{-3}{2}, B = \frac{35}{36}, C = -\frac{3}{2} \log 3 + c$
 c) $A = \frac{5}{2}, B = \frac{31}{36}, C = \frac{3}{2} \log 3 + c$
 d) $A = -\frac{5}{2}, B = \frac{31}{36}, C = \frac{3}{2} \log 3 + c$

Ans: b)

15. $\int \frac{\sin x}{\sin(x-\alpha)} dx =$

- a) $x \cos \alpha + \sin \alpha * \log \sin(x - \alpha) + c$
 b) $x \cos \alpha - \sin \alpha * \log \sin(x - \alpha) + c$
 c) $x \cos \alpha + \sin \alpha * \log \sin(x + \alpha) + c$
 d) $x \cos \alpha - \sin \alpha * \log \sin(x + \alpha) + c$

Ans: a)

16. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be continuous functions, then the value of the integral $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} [f(x) + f(-x)] [g(x) - g(-x)] dx =$

- a) 0
 b) 1
 c) -1

d) 2

Ans: a)

17. If $f(x) = \begin{cases} 2x^2 + 1, & x \leq 1 \\ 4x^2 - 1, & x > 1 \end{cases}$ then $\int_0^2 f(x)dx =$

a) 47/3

b) 50/3

c) 1/3

d) 47/2

Ans: a)

18. The order of the differential equation: $2x^2 \frac{d^2y}{dx^2} - 3 \frac{dy}{dx} + y = 0$ is

a) 2

b) 1

c) 0

d) undefined

Ans: a)

19. The number of arbitrary constants in the general solution of a differential equation of 4th order is

a) 0

b) 2

c) 3

d) 4

Ans: d)

20. The general solution of the differential equation $e^x dy + (ye^x + 2x)dx = 0$ is

a) $xe^x + x^2 = c$

b) $xe^y + y^2 = c$

c) $ye^x + x^2 = c$

d) $xe^x - x^2 = c$

Ans: c)

21. Simplified form of $\cos^{-1}(4x^3 - 3x)$ is

a) $3 \sin^{-1} x$

b) $3 \cos^{-1} x$

c) $\pi - 3 \sin^{-1} x$

d) $\pi - 3 \cos^{-1} x$

Ans: b)

22. For a triangle ΔABC , the value of $b^2 \sin 2C + c^2 \sin 2B$ is

a) Δ

b) 2Δ

c) 3Δ

d) 4Δ

Ans: d)

23. If $\tan \alpha = \frac{a}{a+1}$ and $\tan \beta = \frac{1}{2a+1}$, then $(\alpha + \beta)$ is

a) π

b) $\frac{\pi}{2}$

c) $\frac{\pi}{3}$

d) $\frac{\pi}{4}$

Ans: d)

24. For a triangle ΔABC , the value of $b^2 \sin 2C + c^2 \sin 2B$ is

a) Δ

b) 2Δ

c) 3Δ

d) 4Δ

Ans: d)

25. If $\sin A + \sin B = 2$ then the value of $\sin(A + B)$ is

a) 2

b) 0

c) 1

d) -1

Ans: b)

26.If $\tan 25^\circ = a$ then the value of $\frac{\tan 155^\circ - \tan 115^\circ}{1 + \tan 155^\circ \tan 115^\circ}$ is

- a) $\frac{2a}{1-a^2}$
- b) $\frac{2a}{1+a^2}$
- c) $\frac{1-a^2}{2a}$
- d) $\frac{1+a^2}{2a}$

Ans: c)

27.The length of conjugate axis of a hyperbola is greater than the length of transverse axis. Then the eccentricity e is

- a) $= \sqrt{2}$
- b) $> \sqrt{2}$
- c) $< \sqrt{2}$
- d) $1/\sqrt{2}$

Ans: b)

28.Number of intersecting points of the conic $4x^2 + 9y^2 = 1$ and $4x^2 + y^2 = 4$ is

- a) 1
- b) 2
- c) 3
- d) 0 (zero)

Ans: d)

29.The line segment joining the foci of the hyperbola $x^2 - y^2 + 1 = 0$ is one of the diameters of a circle. The equation of the circle is

$$x^2 + y^2 = 4$$

- a) $x^2 + y^2 = \sqrt{2}$
- b) $x^2 + y^2 = 2$
- c) $x^2 + y^2 = 2\sqrt{2}$

Ans: b)

30. The point Q is the image of the point P (1, 5) about the line $y = x$ and R is the image of the point Q about the line $y = -x$. The circumcentre of the ΔPQR is

- a) (5, 1)
- b) (-5, 1)
- c) (1, -5)
- d) (0, 0)

Ans: d)

31. The foot of the perpendicular drawn from the point (1, 8, 4) on the line joining the points (0, -11, 4) and (2, -3, 1) is

- a) (4, 5, 2)
- b) (-4, 5, 2)
- c) (4, -5, 2)
- d) (4, 5, -2)

Ans: d)

32. If the point of intersection of the lines $2ax + 4ay + c = 0$ and $7bx + 3by - d = 0$ lies in the 4th quadrant and is equidistant from the two axes, where a, b, c and d are non-zero numbers, then ad: bc equals to

- a) 2 : 3
- b) 2 : 1
- c) 1 : 1
- d) 3 : 2

Ans: b)

33. Let $P(n): n^3 + n = 3m$ where m is a positive integer, then which of the following true?

- a) P(1)
- b) P(2)
- c) P(3)
- d) P(4)

Ans: c)

34. Let $n \geq 5; b \neq 0$; if the sum of 5th and 6th terms of the expansion of $(a-b)^n = 0$ then the value of $a/b =$

- a) $5/(n-4)$
- b) $1/5(n-4)$

c) $(n-5)/6$

d) $(n-4)/5$

Ans: d)

35. The coefficient of x^3 in the expansion of 3^x is

a) $\frac{(\log_e 3)^3}{6}$

b) $\frac{(\log_e 3)^3}{3}$

c) $\frac{(\log_e 3)}{6}$

d) $\frac{(\log_e 3)^3}{3}$

Ans: a)

36. If A and B are symmetric matrices then

a) $AB+BA$ is symmetric

b) $AB-BA$ is symmetric

c) $AB-BA$ is skew-symmetric

d) $AB+BA$ is skew-symmetric

Ans: c)

37. If $A = \begin{bmatrix} 3 & 5 \\ 2 & 0 \end{bmatrix}; B = \begin{bmatrix} 1 & 17 \\ 0 & -10 \end{bmatrix}$ then $|AB| =$

a) 80

b) 100

c) 110

d) 92

Ans: b)

38. The roots of the equation $\begin{vmatrix} x & 2 & 1 \\ 2 & 5 & x \\ -1 & 2 & x \end{vmatrix} = 0$ are

a) -3, 1

b) 3, -1

c) 3, 1

d) -3, -1

Ans: b)

39. If $A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$; then A is

- a) involutory matrix
- b) nilpotent matrix of index 2
- c) nilpotent matrix of index 3
- d) rectangular matrix

Ans: c)

40. If one ball is drawn at random from each of the three boxes containing 3 white and 1 black, 2 white and 2 black, 1 white and 3 black balls then the probability that 2 white and 1 black balls will be drawn is

- a) $13/32$
- b) $1/4$
- c) $1/32$
- d) $3/16$

Ans: a)

41. Let A and B two independent events such that their probabilities are $3/10$ and $2/5$. The probability of exactly one of the events happening is

- a) $23/50$
- b) $1/2$
- c) $31/50$
- d) None of these

Ans: a)

42. Let A and B be two sets. Then $(A \cup B)' \cup (A' \cap B)$ is equal to

- a) $A \cup B$
- b) A'
- c) A
- d) None of these

Ans: b)

43. Variance of the data 0,10,20,30,40,50

- a) 291.67
- b) 290
- c) 230
- d) 12

Ans: a)

44. Given set $A = \{1, 2, 3\}$ and a relation $R = \{(1, 2), (2, 1)\}$, the relation R will be

- a) reflexive if $(1, 1)$ is added
- b) symmetric if $(2, 3)$ is added
- c) transitive if $(1, 1)$ is added
- d) symmetric if $(3, 2)$ is added

Ans: c)

45. Let r be the relation over the set $N \times N$ and it is defined by $(a, b) r (c, d) \Rightarrow a + d = b + c$. Then r is

- a) Reflexive only
- b) Symmetric only
- c) Transitive only
- d) An equivalence relation.

Ans: d)

46. The curve $ax + by + xy = 0$ makes an angle $\tan^{-1} 2$ with the X axis at the point $(1, 1)$, then the values of a and b are:

- a) 1, 2
- b) 2, 1
- c) -2, 1
- d) 1, -2

Ans: d)

47. Find the length of the normal at the point $\theta = \pi/2$ on the curve $x = a(\theta + \sin \theta)$, $y = a(1 - \cos \theta)$

- a) $2a$
- b) $a/2$
- c) $a\sqrt{2}$
- d) $a/\sqrt{2}$

Ans: c)

48. The function $f(x) = \cos x - 2kx$ is decreasing when

- a) $k < 2$
- b) $k > 2$
- c) $k > 1/2$
- d) $k < 1/2$

Ans: c)

49. If $\phi(x) = f'(x) + f'(2a - x)$ and $f''(x) > 0, \forall x \in [0, a]$, then the function $\phi(x)$ is

- a) is increasing in the interval $[0, a]$
- b) is increasing in the interval $[0, 2a]$
- c) is decreasing in the interval $[0, a]$
- d) is increasing in the interval $(0, a)$

Ans: c)

50. The area of $\{(x, y) : |x| \geq y \geq x^2\}$:

- a) $2/3$
- b) $1/3$
- c) $1/6$
- d) None of these

Ans: b)

Category- II(Q:51 to 65)

Carry 2 marks each and only one option is correct.

51. If x, y, z are positive numbers in an A.P. then

- a) $\frac{x+y}{2y-x} + \frac{y+z}{2y-z} \geq 4$
- b) $\frac{x+y}{2y-x} + \frac{y+z}{2y-z} \geq 2$
- c) $y^2 \leq zx$
- d) $\frac{x+y}{2y-x} + \frac{y+z}{2y-z} \leq 2$

Ans: a)

52. Let A, G and H be the A.M. , G.M. and H.M. of two numbers a and b .

The quadratic equation whose roots are A and H is:

- a) $Ax^2 - (A^2 + H^2)x + AH^2 = 0$
- b) $Ax^2 - (A^2 + G^2)x + AG^2 = 0$
- c) $Hx^2 - (H^2 + G^2)x + HG^2 = 0$
- d) none of these

Ans: b)

53. If $|z_1|=12$ and $|z_2 - 3 - 4i| = 5$ for all complex numbers z_1, z_2 , then the minimum value of $|z_1 z_2|$ is

- a) 7

- b) 0
- c) 2
- d) 17

Ans: b)

54. The number of seven letter words that can be formed by using the letters of the word SUCCESS so that the two C are together but no two S are together is

- a) 24
- b) 42
- c) 43
- d) 23

Ans: a)

55. Let $f(x) = x|x|$, then the set of points where $f(x)$ is twice differentiable is

- a) $\forall x \in \mathbb{R}$
- b) $\forall x \in \mathbb{R} - \{0\}$
- c) $\forall x \in \mathbb{R} - \{0,1\}$
- d) $\forall x \in \mathbb{R} - \{1\}$

Ans: b)

56. If the function $e^{f(x)} = \frac{10+x}{10-x}$, $x \in (-10,10)$ and $f(x) = kf\left(\frac{200x}{100+x^2}\right)$, then the value of k is

- a) 1/2
- b) 4/5
- c) 7/10
- d) 2/3

Ans: a)

57. If $f: R \rightarrow R$ and $g: R \rightarrow R$ be defined by $f(x) = 2x + 3$, and $g(x) = x^2 + 7$ then the values of x such that $(g \circ f)(x) = 8$ are

- a) 1, -2
- b) -1, 2
- c) -1, -2
- d) 1, 2

Ans: c)

58. The general solution of the differential equation

$$\left(1 + e^{\frac{x}{y}}\right) dx + \left(1 - \frac{x}{y}\right) e^{\frac{x}{y}} dy = 0 \text{ is}$$

a) $x - ye^{\frac{x}{y}} = c$

b) $y - ye^{\frac{x}{y}} = c$

c) $x + ye^{\frac{x}{y}} = c$

d) $y + ye^{\frac{x}{y}} = c$

Ans: c)

59. The possible values of x , which satisfy the trigonometric equation

$$\tan^{-1}\left(\frac{x-1}{x-2}\right) + \tan^{-1}\left(\frac{x+1}{x+2}\right) = \frac{\pi}{4} \text{ are}$$

a) $\pm \frac{1}{\sqrt{2}}$

b) $\pm \sqrt{2}$

c) $\pm \frac{1}{2}$

d) ± 2

Ans: a)

60. If $P = \frac{1}{2}\sin^2\theta + \frac{1}{3}\cos^2\theta$ then

a) $\frac{1}{3} \leq P \leq \frac{1}{2}$

b) $P \leq \frac{1}{2}$

c) $2 \leq P \leq 3$

d) $P \leq \frac{1}{3}$

Ans: a)

61. The length of the chord of the parabola $y^2 = 4ax$ ($a > 0$) which passes through the vertex and makes an acute angle θ with the axis of the parabola is

a) $\pm 4a \cot \theta \operatorname{cosec} \theta$

b) $4a \cot \theta \operatorname{cosec} \theta$

c) $-4a \cot \theta \operatorname{cosec} \theta$

d) $4a \operatorname{cosec}^2 \theta$

Ans: b)

62. If l , m and n (all positive) are p^{th} , q^{th} and r^{th} position terms of any G.P.

series respectively then the value of $\begin{vmatrix} \log l & p & 1 \\ \log m & q & 1 \\ \log n & r & 1 \end{vmatrix} =$

- a) -1
- b) 2
- c) 1
- d) 0

Ans: d)

63. Find the coefficient of $b^{10}a^7c^3$ in the expansion of $(ab+bc+ca)^{10}$

- a) 100
- b) 110
- c) 120
- d) 130

Ans: c)

64. Area of the region bounded by the curves $y = x^2 + 2$, $y = -x$, $x = 0$, $x = 1$ is:

- a) $13/6$
- b) $15/8$
- c) $17/6$
- d) None of these

Ans: c)

65. Let A and B be two independent events such that $P(A) = \frac{1}{5}$, $P(A \cup B) = \frac{7}{10}$. Then $P(\bar{B})$ is equal to

- a) $3/8$
- b) $2/7$
- c) $7/9$
- d) None of these

Ans: a)

Category- III(Q: 66 to 75)

Carry 2 marks each and one or more option(s) is/ are correct.

66. If $x^2 + 4y^2 = 12xy$, $x \in [1,4]$ and $y \in [1,4]$ then

- a) the greatest value of $\log_2(x + 2y)$ is 4
- b) the least value of $\log_2(x + 2y)$ is 3
- c) $\log_2(x + 2y) \in [2,4]$
- d) the number of integral values of (x, y) is 2 such that $\log_2(x + 2y) = 3$

Ans: a), c), d)

67. Let $f(x) = x^2(x + 2) + x + 3$. Then

- a) $f(-3 - k) > 0$ and $f(-2 + k) < 0$ for all $k > 0$
- b) $f(-3 - k) < 0$ and $f(-2 + k) > 0$ for all $k > 0$
- c) $f(x) = 0$ has a root α such that $[\alpha] + 3 = 0$ where $[\alpha]$ is the greatest integer less than or equal to α .
- d) $f(x) = 0$ has a root α such that $(\alpha) + 2 = 0$ where (α) is the smallest integer greater than or equal to α .

Ans: b), c), d)

68. If $z = x + iy$ and the equation $\left| \frac{(2z-i)}{(z+1)} \right| = m$ represents a circle, then m can be

- a) $1/2$
- b) 1
- c) 2
- d) 3

Ans: a), b), d)

69. Number of ways in which three numbers in A.P. can be selected from $1, 2, 3, \dots, n$ is

- a) $\left(\frac{n-1}{2}\right)^2$ if n is even
- b) $\frac{n(n-1)}{4}$ if n is even
- c) $\frac{(n-1)^2}{4}$ if n is odd
- d) none

Ans: b), c)

70. If $f(x) = \begin{cases} x, & \text{when } x \geq 0 \\ -x, & \text{when } x < 0 \end{cases}$ then

- a) $f(x)$ is not differentiable at $x=0$.
- b) $f(x)$ is discontinuous at $x=0$.
- c) $f(x)$ is differentiable at $x=0$.
- d) $f(x)$ is continuous at $x=0$.

Ans: a), d)

71. Let $I_n = \int_0^1 x^n \tan^{-1} x \, dx$. If $a_n I_{n+2} + b_n I_n = c_n$ for all $n \geq 1$, then

- a) a_1, a_2, a_3 are in G. P
- b) b_1, b_2, b_3 are in A. P
- c) c_1, c_2, c_3 are in H. P
- d) a_1, a_2, a_3 are in A. P

Ans: b), d)

72. Let $16x^2 - 3y^2 - 32x - 12y = 44$ represent a hyperbola. Then

- a. Length of the transverse axis is $2\sqrt{3}$
- b. Length of each latus rectum is $32\sqrt{3}$
- c. Eccentricity is $\sqrt{19/3}$
- d. Equation of a directrix is $x = \sqrt{19/3}$

Ans: a), b), c)

73. The line through the points (a, b) and $(-a, -b)$ passes through the point

- a. $(1, 1)$
- b. $(3a, -2b)$
- c. (a^2, ab)
- d. (a, b)

Ans: c), d)

74. If $\Delta(x) = \begin{vmatrix} \sin \frac{x}{2} & 1 & 1 \\ 1 & \sin \frac{x}{2} & -\sin \frac{x}{2} \\ -\sin \frac{x}{2} & 1 & -1 \end{vmatrix}, \forall x \in [0, \pi]$, then

- a. $\Delta(x)$ be maximum at $x = \pi$
- b. $\Delta(x)$ be minimum at $x = 0$
- c. $\Delta(x)$ be maximum at $x=0$ and minimum at $x = \pi$
- d. The range of $\Delta(x)$ be $[2, 4]$

Ans: a), b), d)

75. The vectors $\vec{a} = \alpha\hat{i} + \hat{j} + 2\hat{k}$, $\vec{b} = \hat{i} + \alpha\hat{j} - \hat{k}$ and $\vec{c} = 2\hat{i} - \hat{j} + \alpha\hat{k}$ are coplanar, if

a) $\alpha = -2$

b) $\alpha = \sqrt{3} + 1$

c) $\alpha = 1 - \sqrt{3}$

d) $\alpha = 2$

Ans: a), b), c)